AI-Driven Transformation in Banking GRC: Implications for Workforce Management in UK and EU Banks

A First Review Report

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1 NEED FOR THE STUDY

Banks in the United Kingdom and the European Union are investing substantially in Artificial Intelligence (AI) across their Governance, Risk and Compliance (GRC) functions. AI is now applied to regulatory monitoring, transaction and customer screening, risk scoring, control testing and compliance reporting. The business case for this investment is usually expressed in terms of cost efficiency and faster regulatory response. In practice, however, the returns depend less on the capability of the technology than on the readiness of the workforce that must operate alongside it.

This creates a management problem rather than a technical one. When AI absorbs routine monitoring and reporting work, management must decide which GRC roles are redesigned, which are reduced, and which are newly created; what skills must be built internally or bought in the market; how teams should be structured and staffed; and how employees can be brought to trust and use AI-enabled controls rather than work around them. These are workforce management decisions, and at present they are being taken with very little empirical guidance.

The regulatory environment raises the stakes further. The EU Digital Operational Resilience Act (DORA) and the EU Artificial Intelligence Act impose requirements for operational resilience, risk governance and meaningful human oversight of AI systems. In the United Kingdom, the Prudential Regulation Authority's Supervisory Statement SS1/23 sets expectations for model risk management, and the Senior Managers and Certification Regime (SM&CR) holds named individuals personally accountable for decisions that are increasingly AI-assisted. Staff competence and effective human oversight are therefore not discretionary human resource preferences in this sector; they are conditions of regulatory compliance. A bank that automates a compliance process

without ensuring its people can understand and challenge the output has not reduced its risk but relocated it.

Existing research does not address this intersection. The RegTech literature explains how technology is reshaping regulation and compliance but treats the workforce only implicitly. The work and employment literature establishes how automation recomposes jobs and how employees react to it, but does so across industries rather than within regulated banking functions. The technology acceptance and trust literature explains individual adoption, largely on the basis of experimental studies rather than high-stakes regulated settings.

This study is therefore undertaken to examine AI-driven transformation in UK and EU banking GRC through a workforce management lens, and to develop an actionable framework that banking management can use to plan role redesign, reskilling and employee adaptation while sustaining both compliance effectiveness and employee engagement.

2 REVIEW OF LITERATURE

Artificial Intelligence (AI) in Governance, Risk and Compliance (GRC) has been studied largely as a technological and regulatory development, while the workforce consequences of that development have been examined mainly in general management literature rather than in a banking compliance setting. The following review draws these two streams together.

Davis (1989) established the Technology Acceptance Model, demonstrating through validated measurement scales that an individual's use of a technology is determined by perceived usefulness and perceived ease of use, with perceived usefulness the stronger determinant and ease of use acting largely through it. His work provides the theoretical basis for examining whether compliance and risk professionals will actually rely on AI-enabled controls made available to them.

Autor (2015) examined the history and future of workplace automation and argued that automation substitutes for labour in some tasks while complementing it in others. He showed that machines increase the value of human judgement, problem solving and

adaptability, so that occupations are recomposed rather than eliminated. This distinction between substitution and complementarity underpins the view that AI in GRC leads to role redesign rather than simple job loss.

Cascio and Montealegre (2016) reviewed how technology is changing work and organisations from an organisational psychology and organisational behaviour perspective. They examined the number and types of jobs affected by technological development and the resulting worker dislocation, illustrated the effects through technologies such as electronic monitoring systems and robots, and set out the implications for six human resource areas. Their review positions technological change as a matter of organisational design and human resource management rather than of technology alone.

Frey and Osborne (2017) estimated the probability of computerisation for 702 occupations and concluded that about 47 per cent of total United States employment is at risk, with susceptibility highest for routine and codifiable tasks and negatively related to wages and educational attainment. Their findings are directly relevant to GRC, where regulatory monitoring, transaction screening and compliance reporting involve substantial routine content.

Arner, Barberis and Buckley (2017) introduced the concept of RegTech, arguing that regulatory change and technological development following the 2008 Global Financial Crisis are together transforming regulatory monitoring, reporting and compliance. They contended that RegTech has so far merely digitised manual compliance processes and that its potential lies in near real-time, proportionate regulation, requiring a reconceptualisation of financial regulation at the nexus of data, digital identity and regulation. Their work establishes that compliance work itself is being restructured, not merely supported by technology.

Brougham and Haar (2018) studied employees' perceptions of smart technology, artificial intelligence, robotics and algorithms (STARA) and developed a measure of the extent to which employees believe their job could be replaced by such technologies. Using a mixed-methods design with 120 employees, they found that greater STARA awareness was negatively related to organisational commitment and career satisfaction, and

positively related to turnover intentions, cynicism and depression. Their study demonstrates that the workforce consequences of automation are behavioural and managerial, and not confined to headcount.

Buckley, Arner, Zetzsche and Weber (2020) analysed the European Union's route to RegTech and identified four pillars: extensive post-crisis reporting requirements, strict data protection rules, the facilitation of open banking, and a legislative framework for digital identity. They argued that European financial services and data protection reforms have unintentionally driven the adoption of regulatory technology by intermediaries, supervisors and regulators. Their analysis is important for the present study because it shows regulation acting as a driver of AI adoption in European compliance functions rather than only as a constraint upon it.

Glikson and Woolley (2020) reviewed two decades of empirical research on human trust in AI and concluded that the successful integration of AI into organisations depends critically on workers' trust. They found that cognitive trust is shaped by the tangibility, transparency, reliability and immediacy behaviours of the technology, while emotional trust is associated with anthropomorphism. They also observed that the evidence base relies heavily on short-term, small-sample experimental studies, and noted that trust is likely to develop differently in longer-term, higher-stakes field environments.

This body of work must now be read against a regulatory environment in which workforce capability has become an obligation. The European Union's Digital Operational Resilience Act and Artificial Intelligence Act require operational resilience, risk governance and meaningful human oversight of AI systems, while in the United Kingdom the supervisory expectations on model risk management and the Senior Managers and Certification Regime hold named individuals personally accountable for decisions that are increasingly AI-assisted. Banks in these jurisdictions must therefore be able to demonstrate that their staff are competent to understand, challenge and supervise AI-driven risk and compliance outputs.

Overall, the literature establishes three positions that have not yet been brought together. The RegTech literature explains how compliance functions and regulation are being transformed by technology, but treats the workforce only implicitly. The work and

employment literature establishes that automation recomposes rather than removes jobs, that routine tasks are most exposed, and that employees' perceptions of replaceability carry measurable costs in commitment and turnover, but it is cross-industry rather than specific to banking GRC. The technology acceptance and trust literature identifies the individual-level determinants of adoption, yet, as Glikson and Woolley note, rests largely on experimental settings rather than high-stakes regulated environments. The gap addressed by this study therefore lies at the intersection of the three: how AI-driven transformation of GRC in United Kingdom and European Union banks is reshaping workforce management decisions on role design, skills and staffing, and how employees in a regulated, accountability-bound function adapt to it. The objectives of the study are defined on this basis.

3.1 PRIMARY OBJECTIVE

To examine how AI-driven transformation in the Governance, Risk and Compliance functions of UK and EU banks is reshaping workforce management decisions - role structure, skill requirements, staffing and employee adaptation - and to develop a practical framework enabling banking management to plan GRC workforce transformation while sustaining compliance effectiveness and employee engagement.

3.2 SECONDARY OBJECTIVES

(i) To map which GRC activities are being automated or augmented by AI in UK and EU banks, and how this is changing the demand for compliance and risk roles.

(ii) To analyse how UK and EU regulatory requirements - DORA, the EU AI Act, SS1/23 and SM&CR - shape role design, human oversight and competence obligations within GRC functions.

(iii) To identify emerging skill gaps and reskilling strategies in GRC functions, and to assess how they are being addressed by management.

(iv) To examine employee behavioural responses to AI adoption in GRC - perceived usefulness, trust, perceived replaceability and readiness to adapt - and their implications for adoption and retention.

(v) To develop an integrated workforce transformation framework and recommendations for GRC and HR leadership in UK and EU banks.

4 METHODOLOGY

This study adopts a mixed-methods design, qualitatively dominant, combining analysis of publicly available secondary data with primary data from practitioner interviews and a short follow-up questionnaire. The design reflects both the exploratory nature of the research problem and the practical constraint that data on bank workforce practices is not publicly disclosed at the level of detail required.

4.1 Data Requirement

The study requires four categories of data:

1. **Evidence of AI adoption in GRC functions** - what is being automated or augmented, and in which activities.
2. **Evidence of role and skill change** - how GRC roles, structures and skill requirements are changing.
3. **Regulatory requirements bearing on workforce competence and human oversight** in the UK and EU.
4. **Employee and managerial perceptions** - perceived usefulness of AI in GRC work, trust in AI-generated output, perceived replaceability, skill gaps, and readiness to adapt.

Categories 1 to 3 are obtained from secondary sources; category 4 requires primary data.

4.2 Secondary Data Sources

All secondary sources are publicly available. No confidential, client or employer-specific information is used.

- **Bank disclosures:** annual reports and integrated reports, Pillar 3 disclosures, and operational resilience and technology-risk disclosures of a purposively selected set of eight to ten large UK and EU banks, covering both jurisdictions and a range of business models. Indicative set: HSBC, Barclays, NatWest, Lloyds Banking Group,

Deutsche Bank, BNP Paribas, Banco Santander, ING, UniCredit and Nordea. A common three-year window is used to allow comparison over time.

- **Regulatory and supervisory sources:** Regulation (EU) 2022/2554 (DORA); Regulation (EU) 2024/1689 (EU AI Act); European Banking Authority and European Central Bank publications on the use of AI and on internal governance; PRA Supervisory Statement SS1/23; and FCA publications on AI in UK financial services and on the SM&CR.
- **Labour market data:** publicly advertised GRC, risk, compliance and financial crime job postings from UK and EU banks, collected from public job boards over a defined period, and coded for stated skill requirements. This provides observable evidence of changing skill demand rather than reported opinion, and is used to triangulate the interview findings.
- **Industry surveys and reports:** publicly available surveys on AI adoption in financial services and on compliance workforce trends.

4.3 Research Instruments

Two instruments are used.

(a) Semi-structured interview guide. Approximately ten open-ended questions in four blocks: current use of AI in the respondent's GRC work; changes in tasks and role content; skills and reskilling; and team and individual responses to the change. The guide permits probing and follow-up, which is appropriate given the exploratory objective.

(b) Structured questionnaire. A short instrument of approximately twenty-four items, derived from the themes emerging in the interviews and administered subsequently. Items are measured on a five-point Likert scale and are grouped into constructs adapted from established literature rather than newly invented: perceived usefulness and perceived ease of use (Davis, 1989); STARA awareness and its outcomes of organisational commitment, career satisfaction and turnover intention (Brougham & Haar, 2018); and cognitive trust in AI, operationalised through transparency, reliability and explainability (Glikson & Woolley, 2020). Additional items cover perceived skill gaps, training received and perceived adequacy of human oversight. A short demographic section captures role, function, jurisdiction and years of experience.

Adapting validated constructs allows the questionnaire's reliability to be assessed and permits comparison of the findings with the published literature.

4.4 Sampling Framework

The population is professionals working in or with GRC functions - risk, compliance, internal audit, financial crime, GRC technology and HR business partners supporting these functions - in banks and banking service providers operating in the UK and EU.

Sampling method: purposive sampling, supplemented by snowball referral. Purposive sampling is appropriate because the study requires respondents with direct exposure to AI-enabled GRC processes; a random sample of banking employees would largely comprise respondents unable to answer the questions. Selection criteria are: (i) at least two years' experience in a GRC or GRC-supporting role; (ii) current or recent involvement with a UK or EU banking entity; and (iii) direct or supervisory exposure to AI-enabled or automated GRC processes. Respondents are sought across both jurisdictions, across the three lines of defence, and across both practitioner and managerial levels, so that role-level and decision-level perspectives are both represented.

4.5 Sample Size

- **Interviews:** twelve semi-structured interviews, conducted until thematic saturation, which is conventionally reached in this range for a homogeneous professional group.
- **Questionnaire:** a target of thirty completed responses, treated as a validation of the interview themes rather than as a basis for statistical generalisation.

The sample sizes are deliberately modest and are stated as such: the study's claim is analytical rather than statistical generalisation, and the questionnaire's purpose is to test whether the qualitative themes hold across a wider group of practitioners.

4.6 Tools and Techniques of Analysis

- **Thematic analysis** of interview transcripts, using open coding followed by grouping into themes aligned to the secondary objectives, supported by manual coding in Microsoft Excel.

- **Content analysis** of bank disclosures, regulatory documents and job postings, coding GRC activities as automated, augmented or unaffected, and coding stated skill requirements into categories (technical, analytical, regulatory, behavioural).
- **Descriptive statistics** for questionnaire responses - frequencies, means and standard deviations by construct and by respondent group - using Microsoft Excel or SPSS.
- **Reliability assessment** of the questionnaire constructs using Cronbach's alpha.
- **Cross-tabulation and comparison of group means** between practitioner and managerial respondents, and between UK and EU respondents, to identify differences in perception.
- **Triangulation** of the three evidence streams - disclosures and job postings, interviews, and questionnaire - to test convergence before conclusions are drawn.
- **Framework development**, integrating the validated themes with the reviewed literature to produce the workforce transformation framework.

4.7 Ethical Considerations

Participation is voluntary and informed consent is obtained. All responses are anonymised, and no respondent, employer or client is identifiable in the report. No confidential, proprietary or client data is collected, and respondents are asked to speak to general practice rather than to specific institutional detail. Data is stored securely and used solely for academic purposes.

5. WORK DONE SO FAR

1. **Review of literature completed.** Eight sources have been identified, obtained in full text and reviewed, drawn from management, organisational and financial regulation literature - including *MIS Quarterly*, *Journal of Economic Perspectives*, *Annual Review of Organizational Psychology and Organizational Behavior*, *Technological Forecasting and Social Change*, *Journal of Management & Organization*, *Journal of Banking Regulation* and *Academy of Management Annals*. The review has been written up in narrative form and the research gap identified.

2. **Objectives derived from the literature.** The primary and five secondary objectives have been defined so that each traces to specific reviewed sources.

3. **Research design finalised.** The mixed-methods design, data requirements, sampling framework, sample sizes and analysis tools have been established as set out in Section 4.

4. **Secondary data sources identified.** The set of UK and EU banks, the disclosure types and reporting window, the regulatory sources and the job-posting data source have been identified and confirmed as publicly accessible.

5. **Research instruments drafted.** The semi-structured interview guide and the structured questionnaire have been drafted, with questionnaire constructs adapted from Davis (1989), Brougham & Haar (2018) and Glikson & Woolley (2020).

6 WORK TO BE DONE

1. **Collect the primary data - by early September 2026.** Conduct the twelve semi-structured interviews with the identified respondent group, and circulate the questionnaire, finalised from the interview themes, to obtain the target of thirty responses.

2. **Assemble and analyse the secondary data - by early September 2026.** Complete the content analysis of the selected banks' disclosures, the UK and EU regulatory publications and the GRC job postings.

3. **Analyse the primary data and submit the Second Review report - by 23 September 2026.** Thematically code the interviews, analyse the questionnaire responses using descriptive statistics and reliability assessment, and triangulate all three evidence streams for convergence.

4. **Develop the outputs - during October 2026.** Construct the GRC role-impact map and the workforce transformation framework, and derive the recommendations for GRC and HR leadership.

5. **Complete the project - by 26 October 2026, with the viva voce in November 2026.** Draft and submit the full project report in the prescribed format and prepare for the examination.

7 LIMITATIONS

1. **Sample size and generalisability.** With twelve interviews and approximately thirty questionnaire responses, the study supports analytical rather than statistical generalisation. Findings indicate patterns and mechanisms; they do not establish population-level estimates for UK and EU banking.

2. **Purposive and referral sampling.** Because respondents are selected for their exposure to AI-enabled GRC processes and partly reached through referral, the sample is not random and may over-represent institutions and individuals that are relatively advanced in AI adoption.

3. **Reliance on publicly available data.** Banks do not disclose workforce restructuring, reskilling expenditure or GRC headcount planning in detail. Secondary analysis therefore relies on disclosures prepared for other purposes, and job postings are an indirect proxy for skill demand.

4. **Self-reported perceptions.** Data on trust, perceived usefulness and perceived replaceability is self-reported and may be affected by social desirability bias, particularly where respondents' employers are actively pursuing AI adoption.

5. **Cross-sectional design.** Data is collected at a single point in time, so the study captures perceptions during an ongoing transition and cannot track how roles, skills or attitudes change over time.

6. **A rapidly changing regulatory and technological environment.** The EU AI Act is being phased in and supervisory expectations continue to evolve, so specific regulatory findings are time-bound.

7. **Absence of institutional access.** The study is not conducted as a company project, so no internal bank documentation, policies or workforce data are available; conclusions rest on public sources and practitioner accounts.

8. **Jurisdictional breadth.** The UK and EU are treated as a combined scope, and national differences in labour practice and supervisory approach within the EU are not examined individually.

8 EXPECTED DELIVERABLES

1. An analysis of how AI adoption is changing GRC activities, role structures and skill demand in UK and EU banks.

2. A mapping of UK and EU regulatory requirements - DORA, the EU AI Act, SS1/23 and the SM&CR - to workforce competence and human oversight obligations.

3. A GRC role-impact map classifying activities and roles as automated, augmented or newly created.

4. An assessment of skill gaps and reskilling strategies, including build-versus-buy considerations for GRC talent.

5. An analysis of employee behavioural responses - perceived usefulness, trust, perceived replaceability and readiness to adapt - and their implications for adoption and retention.

6. An integrated workforce transformation framework linking AI adoption in GRC to role redesign, reskilling and employee adaptation.

7. Evidence-based recommendations for GRC and HR leadership in UK and EU banks.

8. The final project report and viva voce presentation.

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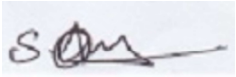
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